Analytic Geometry through Normals, Line Pencils, and Quadratic Forms

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§1.1 Lines as Affine Level Sets

A line in the plane can be written

$$H = \{x \in \mathbb{R}^2 : n \cdot x = c\}, \qquad n \neq 0.$$

The vector $n = (a, b)$ is normal to the line, and a direction vector is

$$d = (-b, a).$$

If $p \in H$, every point of the line has the parametric form

$$x = p + td, \qquad t \in \mathbb{R}.$$

Conversely, eliminating t gives

$$a(x_1 - p_1) + b(x_2 - p_2) = 0.$$

This single description includes vertical lines, where slope notation fails.

The familiar equations are coordinate choices for the same object. If $b \neq 0$,

$$ax + by = c \quad \Longleftrightarrow \quad y = -\frac{a}{b}x + \frac{c}{b}.$$

If a line passes through p_1, p_2, its direction is $p_2 - p_1$. If it meets the coordinate axes away from the origin, dividing by its intercepts produces

$$\frac{x}{a} + \frac{y}{b} = 1.$$

None of these forms is universally best; the normal form survives every orientation, while slope form is efficient only in the nonvertical chart.

In homogeneous coordinates, the line is represented by the coefficient triple

$$\ell = [a : b : -c],$$

and incidence with a projective point $X = [x : y : z]$ is the bilinear equation

$$ax + by - cz = 0.$$

The homogeneous form will later make points at infinity and polar lines ordinary parts of the calculation.

§1.2 Distances and Normal Vectors

Two lines with normals n_1, n_2 are parallel exactly when the normals are proportional. They are perpendicular exactly when

$$n_1 \cdot n_2 = 0,$$

because the direction of each line is a quarter-turn of its normal.

For

$$H = \{x : n \cdot x = c\}$$

and a point x_0, the signed displacement in the normal direction is

$$\frac{n \cdot x_0 - c}{\|n\|_2}.$$

Hence

$$\boxed{\operatorname{dist}_2(x_0, H) = \frac{|n \cdot x_0 - c|}{\|n\|_2}.}$$

To derive it, let

$$p = x_0 - \frac{n \cdot x_0 - c}{\|n\|_2^2} n.$$

Then $n \cdot p = c$, so $p \in H$, and $x_0 - p$ is normal to the line. Any other point of H differs from p by a tangent vector, so the Pythagorean theorem makes p the closest point.

For parallel lines

$$n \cdot x = c_1, \qquad n \cdot x = c_2,$$

the same formula gives

$$\operatorname{dist}_2(H_1, H_2) = \frac{|c_1 - c_2|}{\|n\|_2}.$$

The denominator is not a memorized normalization constant; it compensates for rescaling the defining equation.

§1.3 Hyperplane Distance and Dual Norms

The Euclidean formula has a normed-space version. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$, and let φ be a nonzero linear functional. Its dual norm is

$$\|\varphi\|_* = \sup_{\|u\| \le 1} |\varphi(u)|.$$

For the hyperplane

$$H = \{x : \varphi(x) = c\},$$

one has

$$\boxed{\operatorname{dist}(x_0, H) = \frac{|\varphi(x_0) - c|}{\|\varphi\|_*}.}$$

Indeed, for every $x \in H$,

$$|\varphi(x_0) - c| = |\varphi(x_0 - x)| \le \|\varphi\|_* \|x_0 - x\|,$$

which gives the lower bound. In finite dimensions, the supremum defining the dual norm is attained by a unit vector u; moving from x_0 in the signed u-direction reaches equality.

For $\varphi(x, y) = ax + by$, the dual of ℓ^1 is ℓ^∞, so

$$\operatorname{dist}_1((x_0, y_0), H) = \frac{|ax_0 + by_0 - c|}{\max(|a|, |b|)}.$$

The dual of ℓ^∞ is ℓ^1, giving

$$\operatorname{dist}_\infty((x_0, y_0), H) = \frac{|ax_0 + by_0 - c|}{|a| + |b|}.$$

The familiar square root in the Euclidean formula is therefore a special case of dual-norm geometry.

§1.4 Circles as Quadratic Level Sets

A circle with center $m = (a, b)$ and radius r is

$$\|x - m\|_2^2 = r^2.$$

Expanding gives

$$x^2 + y^2 + Dx + Ey + F = 0,$$

while completing squares recovers

$$m = \left(-\frac{D}{2}, -\frac{E}{2}\right),$$

$$r^2 = \frac{D^2 + E^2 - 4F}{4}.$$

The sign of this final quantity distinguishes a real circle, a point circle, and no real locus.

For a point p, define its power with respect to the circle by

$$\operatorname{Pow}_C(p) = \|p - m\|^2 - r^2.$$

If a line through p meets the circle at a, b, then

$$\operatorname{Pow}_C(p) = \overline{pa}\,\overline{pb}$$

using directed lengths. This follows by substituting the line parametrization $p + td$ into the circle equation: the product of the two parameter roots is the constant term divided by the leading coefficient.

For two circles, the locus of equal powers is their radical axis. Subtracting

$$C_1 : x^2 + y^2 - 4x = 0$$

and

$$C_2 : x^2 + y^2 - 2y - 3 = 0$$

cancels the quadratic terms and gives the line

$$-4x + 2y + 3 = 0.$$

The radical axis is linear because it compares two quadratic functions with the same leading form.

§1.5 Fractional Ranges and Pencils of Lines

Consider

$$\mu = \frac{x}{y-3}$$

under the constraint

$$|x| + |y| \le 1.$$

For fixed μ, rewrite the equation as

$$x = \mu(y-3).$$

This is a line through

$$P = (0,3).$$

The range problem asks which lines in the pencil through P meet the diamond

$$K = \{(x,y) : |x| + |y| \le 1\}.$$

Because K is convex and $P \notin K$, the extreme parameters occur at supporting lines from P to K.

The support function of the ℓ^1-unit ball is

$$h_K(n) = \max_{x \in K} n \cdot x = \|n\|_\infty.$$

The line can be written

$$(1, -\mu) \cdot (x,y) = -3\mu.$$

It meets K precisely when

$$|3\mu| \le h_K(1,-\mu) = \max(1, |\mu|).$$

If $|\mu| > 1$, this is impossible. If $|\mu| \le 1$, it becomes

$$3|\mu| \le 1.$$

Therefore

$$\boxed{\mu \in \left[-\frac{1}{3}, \frac{1}{3}\right].}$$

The endpoints correspond to the supporting lines through the vertices $(1,0)$ and $(-1,0)$. Convex duality has not replaced the elementary picture; it has explained why the endpoints are support data.

§1.6 Ratios on a graph are radial coordinates

For a point $(x, f(x))$ with $x \neq 0$, the ratio

$$\frac{f(x)}{x}$$

is the slope of the ray from the origin to the graph point. Counting solutions of

$$\frac{f(x)}{x} = m$$

is therefore the same as counting intersections between the graph and the line

$$y = mx.$$

A rational expression has become a radial-projection problem.

This viewpoint also identifies tangency thresholds. As m varies, the number of intersections can change only when the line becomes tangent to the graph or passes through a singular point. Algebraically, solving

$$f(x) = mx$$

together with

$$f'(x) = m$$

detects those critical slopes. The graphical and derivative descriptions are the same incidence event viewed globally and locally.

§1.7 The Moving Line and Its Circle

Let fixed points be

$$A = (-3, 0), \qquad B = (1, 6).$$

Suppose lines through A and B meet at P and are perpendicular. Then

$$\angle APB = 90^\circ,$$

so Thales' theorem places P on the circle with diameter AB.

The right triangle gives

$$PA^2 + PB^2 = AB^2.$$

Hence

$$PA\,PB \le \frac{PA^2 + PB^2}{2} = \frac{AB^2}{2}.$$

Since

$$AB^2 = (1+3)^2 + 6^2 = 52,$$

one obtains

$$\boxed{PA\,PB \le 26.}$$

Equality occurs when $PA = PB$, namely at the intersections of the diameter circle with the perpendicular bisector of AB.

The coordinate version would introduce two slopes and a quadratic constraint. The circle formulation identifies the locus first, after which the optimization is a one-line inequality.

§1.8 A general conic is an affine quadratic form

Write an affine conic as

$$F(x) = x^T Q x + 2\ell^T x + f = 0, \qquad Q = Q^T.$$

An orthogonal change of coordinates diagonalizes Q, removing the cross term without changing Euclidean angles. If Q is invertible, translating by

$$x = u - Q^{-1}\ell$$

removes the linear term. Rotation handles directions; translation handles the center.

Consider

$$xy - x - y = 0.$$

Set

$$u = \frac{x+y}{\sqrt{2}}, \qquad v = \frac{x-y}{\sqrt{2}}.$$

Then

$$xy = \frac{u^2 - v^2}{2}, \qquad x + y = \sqrt{2}\, u.$$

The equation becomes

$$u^2 - v^2 - 2\sqrt{2}\, u = 0,$$

or

$$\boxed{(u - \sqrt{2})^2 - v^2 = 2.}$$

The original rectangular-looking equation is a hyperbola whose principal axes are the eigenvectors of the quadratic matrix

$$Q = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

§1.9 Rank and the line at infinity distinguish conic types

Homogenizing a conic gives

$$X^T A X = 0, \qquad X = (x, y, z)^T,$$

for a symmetric 3×3 matrix A. The conic is degenerate when A loses rank. For example,

$$xy = 0$$

is the union of two lines and has a rank-deficient homogeneous matrix.

The intersection with the line at infinity $z = 0$ records affine directions.

- An ellipse
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2$$
has no real point at infinity.
- A parabola
$$y^2 = 2pxz$$
meets the line at infinity at one double point $[1 : 0 : 0]$.
- A hyperbola
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z^2$$
has two real points at infinity, corresponding to its asymptotic directions.

For

$$xy - x - y = 0,$$

the projective closure is

$$XY - XZ - YZ = 0.$$

At infinity,

$$Z = 0 \quad \Longrightarrow \quad XY = 0,$$

so the two points are

$$[1 : 0 : 0], \qquad [0 : 1 : 0].$$

They are the projective endpoints of the two asymptote directions revealed by the standard hyperbola form.

§1.10 Conic Polarity and Tangents

Let a nondegenerate projective conic be

$$F(X) = X^T AX = 0, \qquad A = A^T.$$

For a point p, its polar line is

$$p^T AX = 0.$$

If p lies on the conic, this polar is the tangent line. Indeed, the differential is

$$dF_p(H) = 2p^T AH.$$

The tangent condition at p is

$$p^T A(X - p) = 0.$$

Since $p^T Ap = 0$, this reduces to

$$p^T AX = 0.$$

For the conic

$$XY - XZ - YZ = 0,$$

use the symmetric matrix

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix},$$

which represents twice the equation. The affine point $(2, 2)$ corresponds to

$$p = [2 : 2 : 1]$$

and lies on the conic. Since

$$Ap = \begin{pmatrix} 1 \\ 1 \\ -4 \end{pmatrix},$$

its polar, hence its tangent, is

$$X + Y - 4Z = 0.$$

In the affine chart $Z = 1$,

$$\boxed{x + y = 4.}$$

The same answer follows from the gradient of $xy-x-y$ at $(2, 2)$, but the polar calculation remains valid in homogeneous coordinates and transforms naturally with the conic under projective changes of coordinates.